

EEG SIGNAL PROCESSING AND SPECTRAL ENTROPY ANALYSIS

1. Introduction

The aim of this analysis is to measure the depth of anesthesia using a feature extracted from the EEG signal, specifically the complexity of the spectrum. Spectral entropy is used as a measure of signal complexity, providing a quantitative index that is high when the patient is awake and low during deep anesthesia.

This index is compared with the commercial Bispectral Index (BIS) monitor, which outputs values between 0 (deep anesthesia) and 100 (fully awake). The EEG data were recorded during sevoflurane anesthesia and include a single EEG channel, annotations marking the periods before and after anesthesia, and the BIS monitor signal.

2. Methods

Before processing, the EEG signal was pre-filtered to remove very low frequencies that typically occur due to movement artifacts. A filter suppresses unwanted components such as noise or interference while allowing the desired signal to pass. (Seppälä, 2026, p. 39).

Two types of filters were considered: FIR (finite impulse response), which depend on current and past inputs, and IIR (infinite impulse response), which also uses past outputs. (s.32). FIR filters provide linear phase response while IIR filters require lower order but can be less stable and may distort phase. (Seppälä, 2026, p. 37).

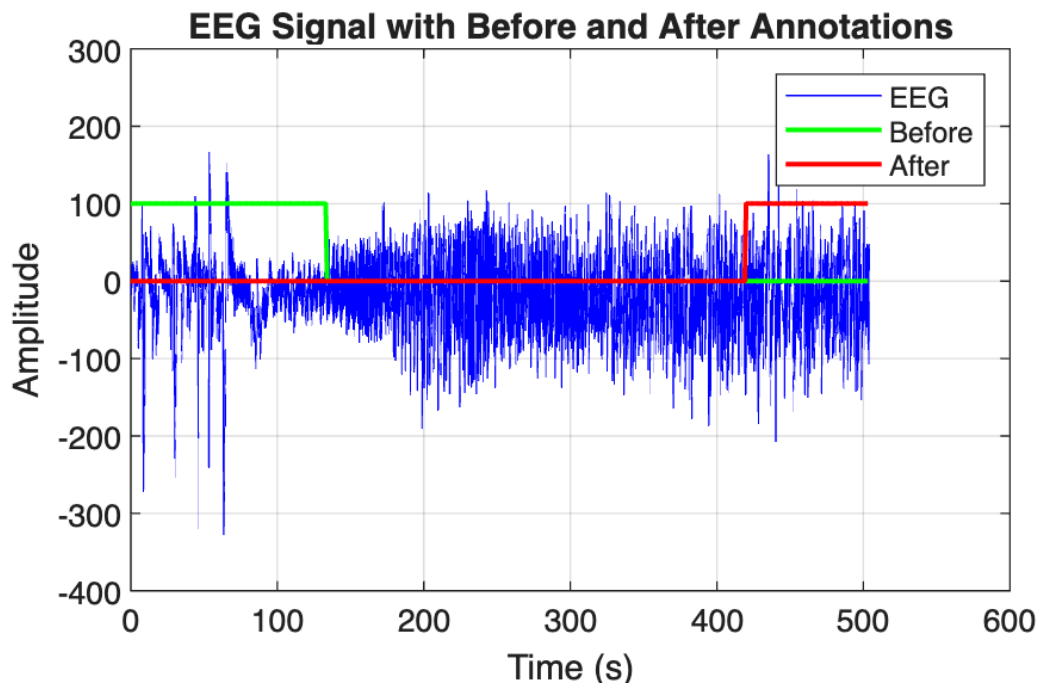
After testing both filters, the FIR filter was preferred because it is always stable and preserves the waveform shape of the EEG signal. Practical filters cannot completely remove noise without affecting the signal, since signal and noise often share frequency ranges (Seppälä, 2026, p. 38).

The filtered EEG signal was divided into 5-second windows with 50% overlap using the provided entropy function. For each window, the power spectral density (PSD) was computed using a spectral transformation. The PSD was then normalized to form a probability distribution, after which spectral entropy was calculated based on Shannon

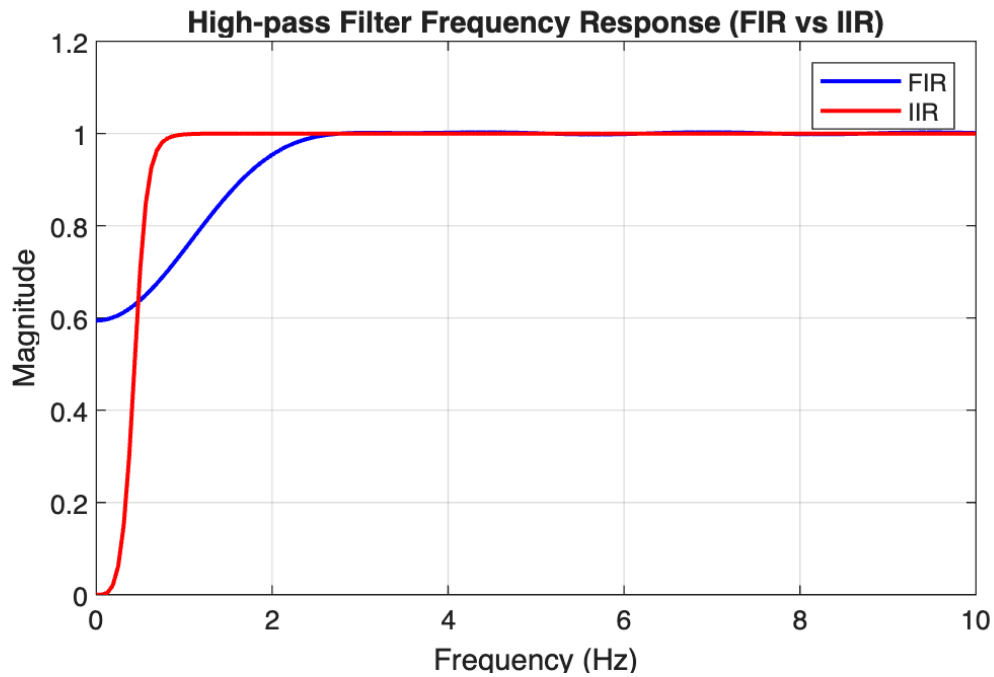
entropy. Finally, the entropy values were normalized by the logarithm of the number of frequency components, resulting in values between 0 and 1, and scaled to 0-100 for easier interpretation.

3. Result and Discussion

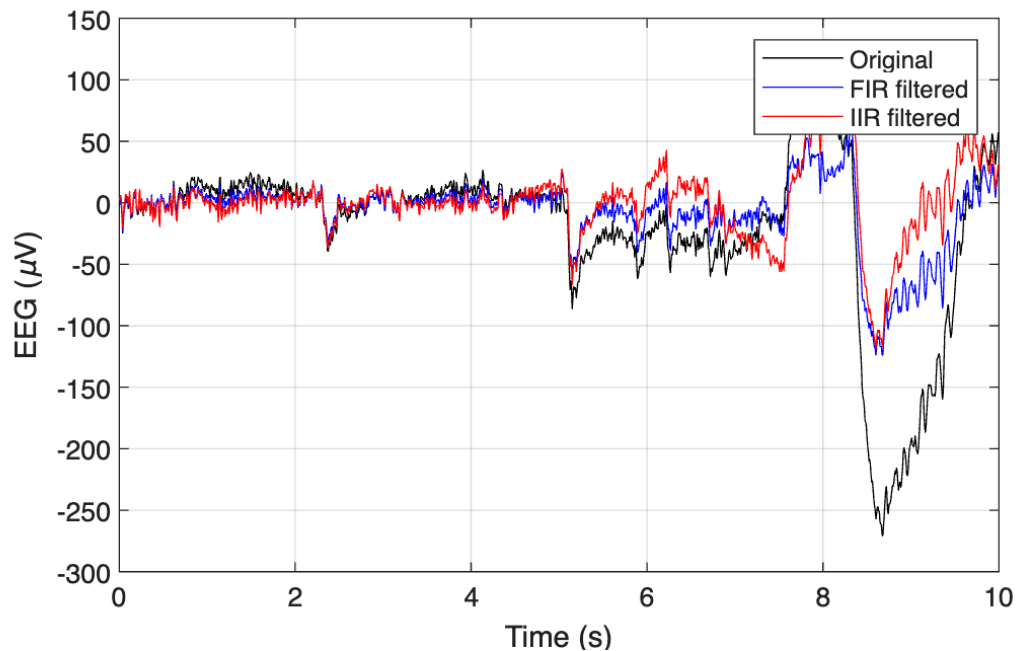
The first tasks involved plotting the EEG signal together with the ‘before’ and ‘after’ anesthesia annotations. The EEG sampling frequency was 128 Hz. The combined plot shows that during anesthesia, the EEG more slower and more regular, with reduced high-frequency activity, compared to the awake state. At the start of the recording, the ‘before’ annotation is high, indicating the patient is awake, and the EEG appears more irregular, with higher-frequency activity. In the middle of the recording, both annotation signals are zero, corresponding to the anesthesia period, during which the EEG becomes slower and more regular. Toward the end of the recording, the ‘after’ signal increases while the ‘before’ signal remains zero, indicating recovery, and the EEG returns to a more irregular pattern resembling the awake state.



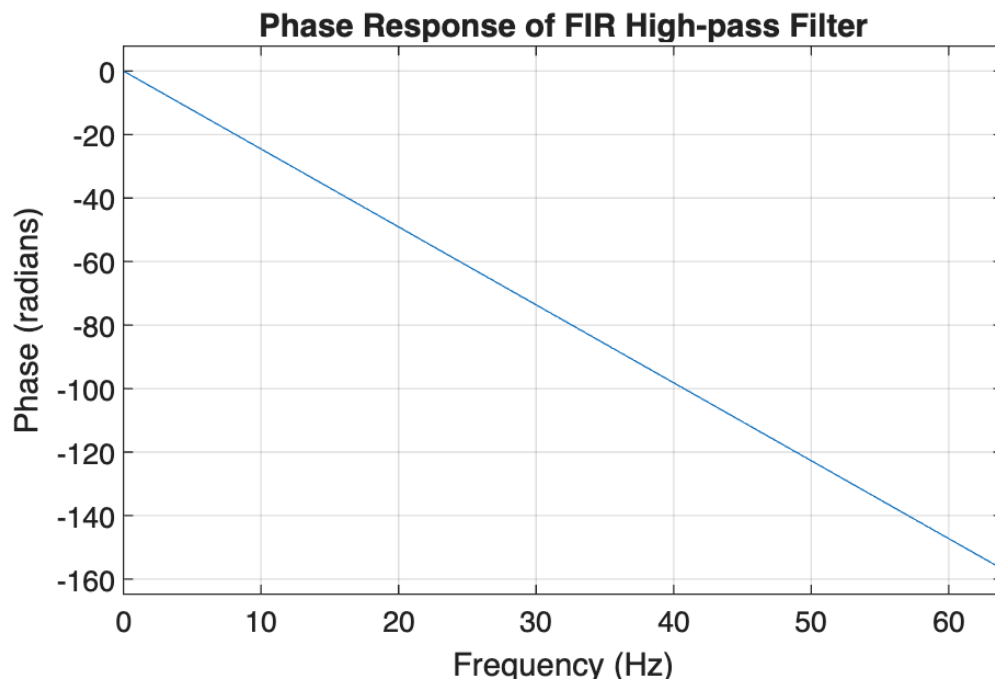
Both FIR and IIR high-pass filters effectively remove slow baseline below 0.5 Hz. The FIR filter shows a smooth transition with linear phase, whereas the IIR filter has a sharper cutoff. While both filters perform effectively, the FIR was preferred because it is always stable, while the IIR filter can be less stable under certain conditions (Seppälä, 2025, p. 4).



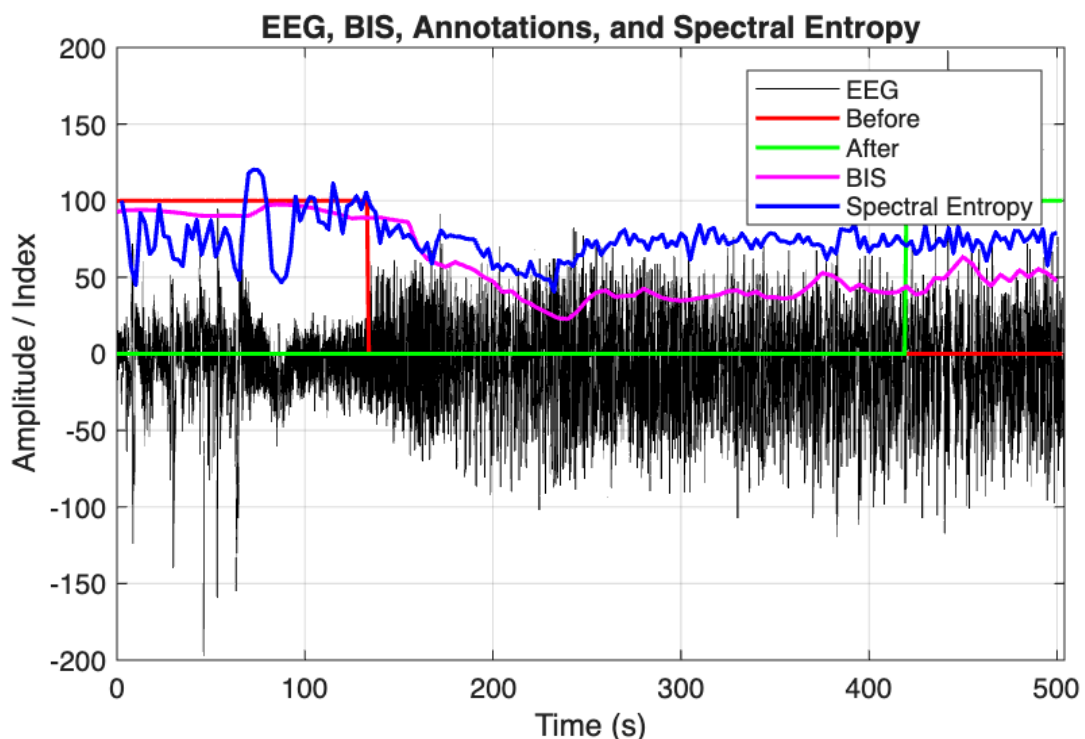
In the next filtered EEG plot, the blue line representing the FIR-filtered EEG aligns more closely with the original EEG (black) better than the red line representing the IIR-filtered EEG. This indicates that the FIR filter preserves the waveform shape and peak timing more accurately while effectively removing slow baseline drift.



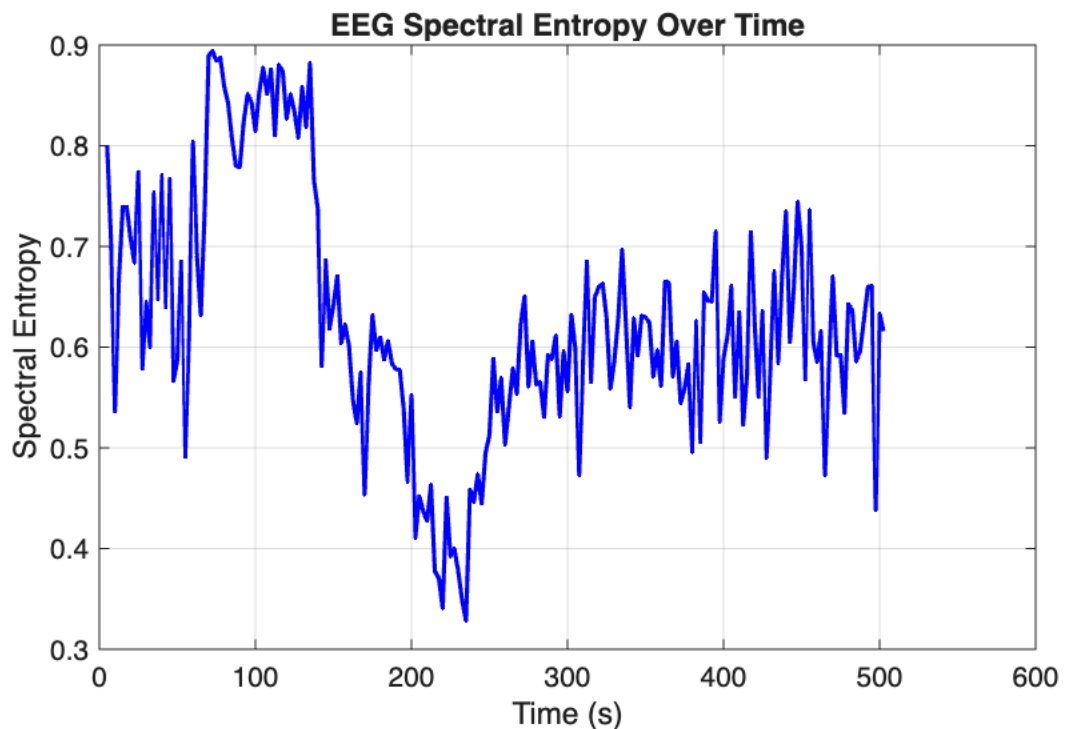
The phase response of the FIR high-pass filter was examined in the next figure. The FIR phase response is approximately a straight line from 0 to -160 degrees, indicating linear phase. This means that all frequency components of the EEG signal are delayed equally.



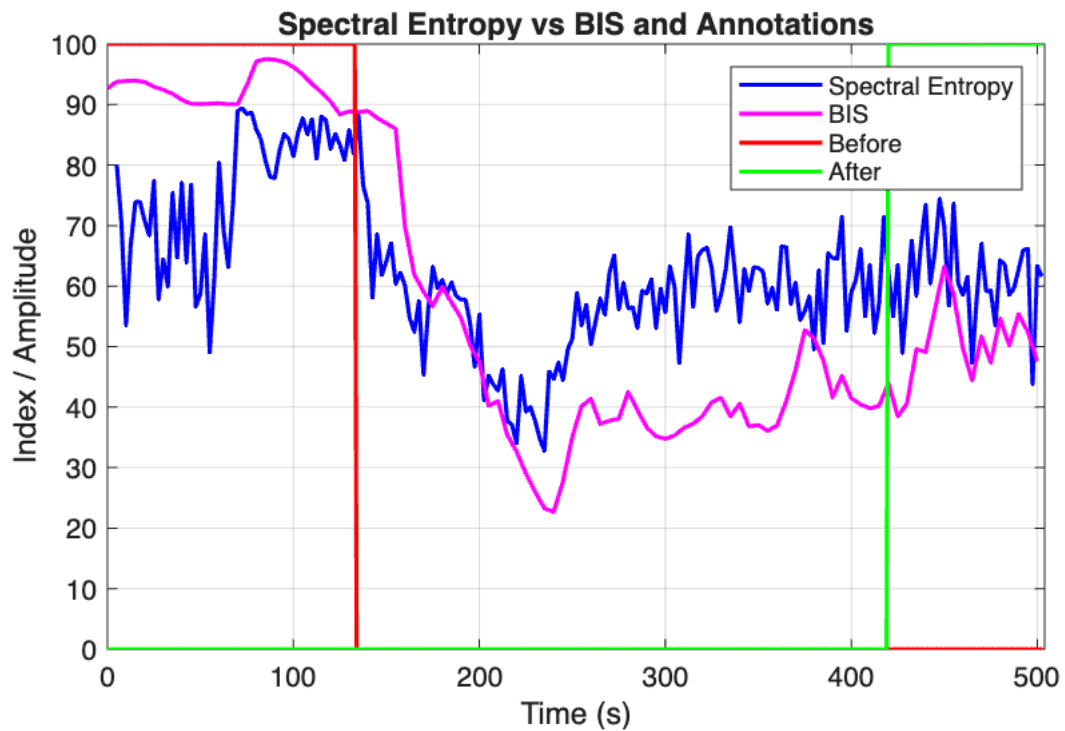
As shown in the next figure, spectral entropy exhibits high values during the ‘before’ phase, indicating an awake state with complex EEG activity. Around 140 seconds, a clear drop in entropy corresponds to the transition to anesthesia. During the ‘after’ phase, entropy stabilizes at lower values (50-70), reflecting reduced spectral complexity. The trend closely follows the BIS signal, with both measures decreasing during induction and remaining low during anesthesia, although spectral entropy exhibits more variability.



The spectral entropy of the EEG reflects the complexity of brain activity. Higher values (0.8-0.9) indicate awake states, while lower values (0.35-0.5) correspond to deeper anesthesia. The plot below shows a clear decrease in entropy during anesthesia, demonstrating its usefulness for monitoring anesthesia depth.



From the next figure which shows the scaled spectral entropy (0-100), BIS index, and the before/after annotations over time. Both spectral entropy and BIS decrease during anesthesia, reflecting reduced brain activity complexity. Their general trends align closely, although not perfectly. The spectral entropy signal is noisier compared to BIS index. These differences likely arise because the BIS employs advanced processing and artifact rejection, whereas spectral entropy is a direct measure of EEG spectral complexity.



4. Conclusion

In this analysis, the depth of anesthesia was evaluated using EEG spectral entropy as a measure of brain activity complexity. The EEG signal was first preprocessed with a FIR high-pass filter to remove slow baseline drifts and then divided into overlapping 5- second windows for power spectral density estimation. Spectral entropy was calculated for each window and scaled to 0-100 for easier interpretation.

The results demonstrate that spectral entropy reflects the patient's state: higher values correspond to awake periods, while lower values indicate deep anesthesia. Comparison with the BIS index shows that spectral entropy follows the same overall trend, although it is more variable due to the direct calculation from EEG without artifact suppression.

References

Seppälä, S. (2025). *Assignment 2: Processing of Biosignals* [Course material]. Tampere University.

Seppälä, S. (2026). *Processing of biosignals: Lecture32 – Noise in Biosignals and Filtering concepts*. Tampere University of Technology. Course lecture material.

